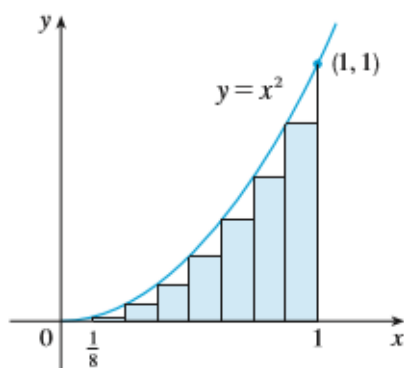


Integrals

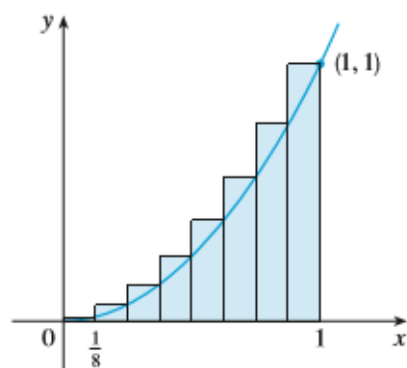
4.1 Areas and Distances

The Area Problem

EXAMPLE Use rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1



(a) Using left endpoints



(b) Using right endpoints

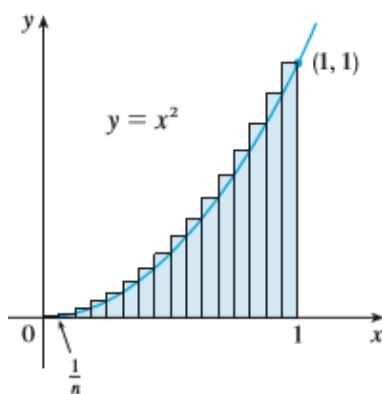
By computing the sum of the areas of the smaller rectangles (L_8) and the sum of the areas of the larger rectangles (R_8), we obtain better lower and upper estimates for A :

$$0.2734375 < A < 0.3984375$$

EXAMPLE For the region S in Example 1, show that the sum of the areas of the upper approximating rectangles approaches $\frac{1}{3}$, that is,

$$\lim_{n \rightarrow \infty} R_n = \frac{1}{3}$$

SOLUTION R_n is the sum of the areas of the n rectangles in Figure 7. Each rectangle has width $1/n$ and the heights are the values of the function $f(x) = x^2$ at the points $1/n, 2/n, 3/n, \dots, n/n$; that is, the heights are $(1/n)^2, (2/n)^2, (3/n)^2, \dots, (n/n)^2$. Thus



$$\begin{aligned} R_n &= \frac{1}{n} \left(\frac{1}{n} \right)^2 + \frac{1}{n} \left(\frac{2}{n} \right)^2 + \frac{1}{n} \left(\frac{3}{n} \right)^2 + \cdots + \frac{1}{n} \left(\frac{n}{n} \right)^2 \\ &= \frac{1}{n} \cdot \frac{1}{n^2} (1^2 + 2^2 + 3^2 + \cdots + n^2) \\ &= \frac{1}{n^3} (1^2 + 2^2 + 3^2 + \cdots + n^2) \end{aligned}$$

Here we need the formula for the sum of the squares of the first n positive integers:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Definition The **area** A of the region S that lies under the graph of the continuous function f is the limit of the sum of the areas of approximating rectangles:

$$A = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} [f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x]$$

The Distance Problem

$$\text{distance} = \text{velocity} \times \text{time}$$

In general, suppose an object moves with velocity $v = f(t)$, where $a \leq t \leq b$ and $f(t) \geq 0$ (so the object always moves in the positive direction). We take velocity readings at times $t_0 (= a)$, $t_1, t_2, \dots, t_n (= b)$ so that the velocity is approximately constant on each subinterval. If these times are equally spaced, then the time between consecutive readings is $\Delta t = (b - a)/n$. During the first time interval the velocity is approximately $f(t_0)$ and so the distance traveled is approximately $f(t_0) \Delta t$. Similarly, the distance traveled during the second time interval is about $f(t_1) \Delta t$ and the total distance traveled during the time interval $[a, b]$ is approximately

$$f(t_0) \Delta t + f(t_1) \Delta t + \cdots + f(t_{n-1}) \Delta t = \sum_{i=1}^n f(t_{i-1}) \Delta t$$

If we use the velocity at right endpoints instead of left endpoints, our estimate for the total distance becomes

$$f(t_1) \Delta t + f(t_2) \Delta t + \cdots + f(t_n) \Delta t = \sum_{i=1}^n f(t_i) \Delta t$$

the *exact* distance d traveled is the *limit* of such expressions

$$d = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_{i-1}) \Delta t = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_i) \Delta t$$

4.2 The Definite Integral

Definition of a Definite Integral If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0 (= a)$, $x_1, x_2, \dots, x_n (= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any **sample points** in these subintervals, so x_i^* lies in the i th subinterval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is **integrable** on $[a, b]$.

NOTE 1 The symbol $\int$ was introduced by Leibniz and is called an **integral sign**. It is an elongated S and was chosen because an integral is a limit of sums. In the notation $\int_a^b f(x) dx$, $f(x)$ is called the **integrand** and a and b are called the **limits of integration**; a is the **lower limit** and b is the **upper limit**. For now, the symbol dx has no meaning by itself; $\int_a^b f(x) dx$ is all one symbol. The dx simply indicates that the independent variable is x . The procedure of calculating an integral is called **integration**.

NOTE 2 The definite integral $\int_a^b f(x) dx$ is a number; it does not depend on x . In fact, we could use any letter in place of x without changing the value of the integral:

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(r) dr$$

NOTE 3 The sum

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

that occurs in Definition 2 is called a **Riemann sum** after the German mathematician Bernhard Riemann (1826–1866).

Theorem If f is continuous on $[a, b]$, or if f has only a finite number of jump discontinuities, then f is integrable on $[a, b]$; that is, the definite integral $\int_a^b f(x) dx$ exists.

Theorem If f is integrable on $[a, b]$, then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

where $\Delta x = \frac{b-a}{n}$ and $x_i = a + i \Delta x$

Midpoint Rule

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(\bar{x}_i) \Delta x = \Delta x [f(\bar{x}_1) + \cdots + f(\bar{x}_n)]$$

where $\Delta x = \frac{b-a}{n}$

and $\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i]$

EXAMPLE Use the Midpoint Rule with $n = 5$ to approximate $\int_1^2 \frac{1}{x} dx$.

Properties of the Definite Integral

$$\int_b^a f(x) dx = -\int_a^b f(x) dx$$

If $a = b$, then $\Delta x = 0$ and so

$$\int_a^a f(x) dx = 0$$

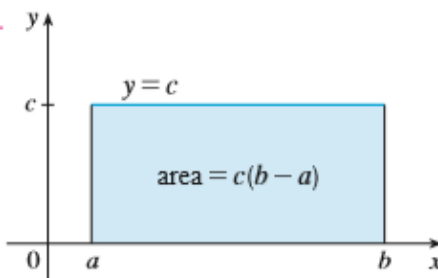
Properties of the Integral

1. $\int_a^b c dx = c(b-a)$, where c is any constant

2. $\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$

3. $\int_a^b c f(x) dx = c \int_a^b f(x) dx$, where c is any constant

4. $\int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$



5. $\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$

Comparison Properties of the Integral

6. If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq 0$.
7. If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$.
8. If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$$

4.3 The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus, Part 1 If f is continuous on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t) dt \quad a \leq x \leq b$$

is continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) = f(x)$.

PROOF If x and $x + h$ are in (a, b) , then

$$\begin{aligned} g(x + h) - g(x) &= \int_a^{x+h} f(t) dt - \int_a^x f(t) dt \\ &= \left(\int_a^x f(t) dt + \int_x^{x+h} f(t) dt \right) - \int_a^x f(t) dt \quad (\text{by Property 5}) \\ &= \int_x^{x+h} f(t) dt \end{aligned}$$

and so, for $h \neq 0$,

$$\boxed{2} \quad \frac{g(x + h) - g(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt$$

For now let's assume that $h > 0$. Since f is continuous on $[x, x + h]$, the Extreme Value Theorem says that there are numbers u and v in $[x, x + h]$ such that $f(u) = m$ and $f(v) = M$, where m and M are the absolute minimum and maximum values of f on $[x, x + h]$. (See Figure 6.)

By Property 8 of integrals, we have

$$mh \leq \int_x^{x+h} f(t) dt \leq Mh$$

that is,

$$f(u)h \leq \int_x^{x+h} f(t) dt \leq f(v)h$$

Since $h > 0$, we can divide this inequality by h :

$$f(u) \leq \frac{1}{h} \int_x^{x+h} f(t) dt \leq f(v)$$

Now we use Equation 2 to replace the middle part of this inequality:

$$\boxed{3} \quad f(u) \leq \frac{g(x+h) - g(x)}{h} \leq f(v)$$

Inequality 3 can be proved in a similar manner for the case where $h < 0$. (See Exercise 71.)

Now we let $h \rightarrow 0$. Then $u \rightarrow x$ and $v \rightarrow x$, since u and v lie between x and $x+h$. Therefore

$$\lim_{h \rightarrow 0} f(u) = \lim_{u \rightarrow x} f(u) = f(x) \quad \text{and} \quad \lim_{h \rightarrow 0} f(v) = \lim_{v \rightarrow x} f(v) = f(x)$$

because f is continuous at x . We conclude, from $\boxed{3}$ and the Squeeze Theorem, that

$$\boxed{4} \quad g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = f(x)$$

If $x = a$ or b , then Equation 4 can be interpreted as a one-sided limit. Then Theorem 2.8.4 (modified for one-sided limits) shows that g is continuous on $[a, b]$. ■

Using Leibniz notation for derivatives, we can write FTC1 as

$$\boxed{5} \quad \frac{d}{dx} \int_a^x f(t) dt = f(x)$$

when f is continuous. Roughly speaking, Equation 5 says that if we first integrate f and then differentiate the result, we get back to the original function f .

The Fundamental Theorem of Calculus, Part 2 If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function such that $F' = f$.

PROOF Let $g(x) = \int_a^x f(t) dt$. We know from Part 1 that $g'(x) = f(x)$; that is, g is an antiderivative of f . If F is any other antiderivative of f on $[a, b]$, then we know from Corollary 4.2.7 that F and g differ by a constant:

$$\boxed{6} \quad F(x) = g(x) + C$$

for $a < x < b$. But both F and g are continuous on $[a, b]$ and so, by taking limits of both sides of Equation 6 (as $x \rightarrow a^+$ and $x \rightarrow b^-$), we see that it also holds when $x = a$ and $x = b$.

If we put $x = a$ in the formula for $g(x)$, we get

$$g(a) = \int_a^a f(t) dt = 0$$

So, using Equation 6 with $x = b$ and $x = a$, we have

$$\begin{aligned} F(b) - F(a) &= [g(b) + C] - [g(a) + C] \\ &= g(b) - g(a) = g(b) = \int_a^b f(t) dt \end{aligned}$$

Differentiation and Integration as Inverse Processes

bring together the two parts of the Fundamental Theorem.

The Fundamental Theorem of Calculus Suppose f is continuous on $[a, b]$.

1. If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$.
2. $\int_a^b f(x) dx = F(b) - F(a)$, where F is any antiderivative of f , that is, $F' = f$.

4.4 Indefinite Integrals and the Net Change Theorem

Indefinite Integrals

We need a convenient notation for antiderivatives that makes them easy to work with. Because of the relation given by the Fundamental Theorem between antiderivatives and integrals, the notation $\int f(x) dx$ is traditionally used for an antiderivative of f and is called an **indefinite integral**. Thus

$$\int f(x) dx = F(x) \quad \text{means} \quad F'(x) = f(x)$$

You should distinguish carefully between definite and indefinite integrals. A definite integral $\int_a^b f(x) dx$ is a *number*; whereas an indefinite integral $\int f(x) dx$ is a *function* (or family of functions). The connection between them is given by Part 2 of the Fundamental Theorem: If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = \left[\int f(x) dx \right]_a^b$$

1 Table of Indefinite Integrals

$$\int cf(x) dx = c \int f(x) dx$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int k dx = kx + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{1}{x^2 + 1} dx = \tan^{-1} x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$

Applications

Net Change Theorem The integral of a rate of change is the net change:

$$\int_a^b F'(x) dx = F(b) - F(a)$$

WRITING PROJECT

NEWTON, LEIBNIZ, AND THE INVENTION OF CALCULUS

We sometimes read that the inventors of calculus were Sir Isaac Newton (1642–1727) and Gottfried Wilhelm Leibniz (1646–1716). But we know that the basic ideas behind integration were investigated 2500 years ago by ancient Greeks such as Eudoxus and Archimedes, and methods for finding tangents were pioneered by Pierre Fermat (1601–1665), Isaac Barrow (1630–1677), and others.

Read about the contributions of these men in one or more of the given references and write a report on one of the following three topics. You can include biographical details, but the main thrust of your report should be a description, in some detail, of their methods and notations. In particular, you should consult one of the sourcebooks, which give excerpts from the original publications of Newton and Leibniz, translated from Latin to English.

- The Role of Newton in the Development of Calculus
- The Role of Leibniz in the Development of Calculus
- The Controversy between the Followers of Newton and Leibniz over Priority in the Invention of Calculus

4.5 The Substitution Rule

The Substitution Rule If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x))g'(x)dx = \int f(u)du$$

EXAMPLE Find $\int x^3 \cos(x^4 + 2) dx$.

SOLUTION We make the substitution $u = x^4 + 2$

EXAMPLE Calculate $\int \tan x \, dx$.

SOLUTION First we write tangent in terms of sine and cosine:

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$$

This suggests that we should substitute $u = \cos x$, since then $du = -\sin x \, dx$ and so $\sin x \, dx = -du$:

$$\begin{aligned} \int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx = -\int \frac{1}{u} \, du \\ &= -\ln |u| + C = -\ln |\cos x| + C \end{aligned}$$

Since $-\ln |\cos x| = \ln(|\cos x|^{-1}) = \ln(1/|\cos x|) = \ln |\sec x|$, the result of Example 6 can also be written as

$$\int \tan x \, dx = \ln |\sec x| + C$$

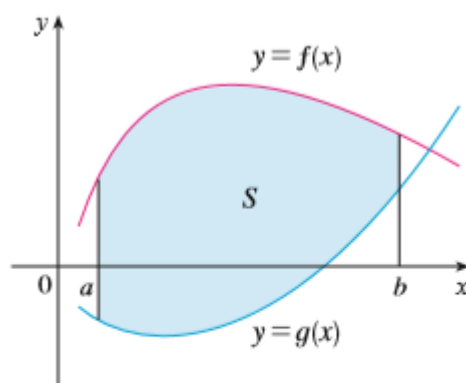
The Substitution Rule for Definite Integrals If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x)) g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du$$

Integrals of Symmetric Functions Suppose f is continuous on $[-a, a]$.

- (a) If f is even [$f(-x) = f(x)$], then $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$.
- (b) If f is odd [$f(-x) = -f(x)$], then $\int_{-a}^a f(x) \, dx = 0$.

5.1 Areas Between Curves



FIGURE

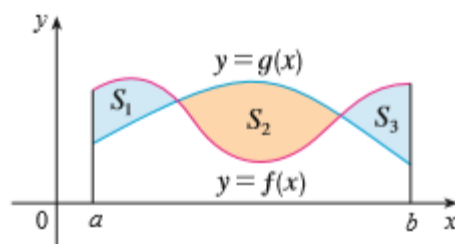
$$S = \{(x, y) \mid a \leq x \leq b, g(x) \leq y \leq f(x)\}$$

The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, is

$$A = \int_a^b [f(x) - g(x)] dx$$

.....

If we are asked to find the area between the curves $y = f(x)$ and $y = g(x)$ where $f(x) \geq g(x)$ for some values of x but $g(x) \geq f(x)$ for other values of x



The area between the curves $y = f(x)$ and $y = g(x)$ and between $x = a$ and $x = b$ is

$$A = \int_a^b |f(x) - g(x)| dx$$

APPLIED PROJECT

THE GINI INDEX

How is it possible to measure the distribution of income among the inhabitants of a given country? One such measure is the *Gini index*, named after the Italian economist Corrado Gini who first devised it in 1912.

We first rank all households in a country by income and then we compute the percentage of households whose income is at most a given percentage of the country's total income. We define a **Lorenz curve** $y = L(x)$ on the interval $[0, 1]$ by plotting the point $(a/100, b/100)$ on the curve if the bottom $a\%$ of households receive at most $b\%$ of the total income. For instance, in Figure 1 the point $(0.4, 0.12)$ is on the Lorenz curve for the United States in 2008 because the poorest 40% of the population received just 12% of the total income. Likewise, the bottom 80% of the population received 50% of the total income, so the point $(0.8, 0.5)$ lies on the Lorenz curve. (The Lorenz curve is named after the American economist Max Lorenz.)

Figure 2 shows some typical Lorenz curves. They all pass through the points $(0, 0)$ and $(1, 1)$ and are concave upward. In the extreme case $L(x) = x$, society is perfectly egalitarian: The poorest $a\%$ of the population receives $a\%$ of the total income and so everybody receives the same income. The area between a Lorenz curve $y = L(x)$ and the line $y = x$ measures how much the income distribution differs from absolute equality. The **Gini index** (sometimes called the **Gini coefficient** or the **coefficient of inequality**) is the area between the Lorenz curve and the line $y = x$ (shaded in Figure 3) divided by the area under $y = x$.

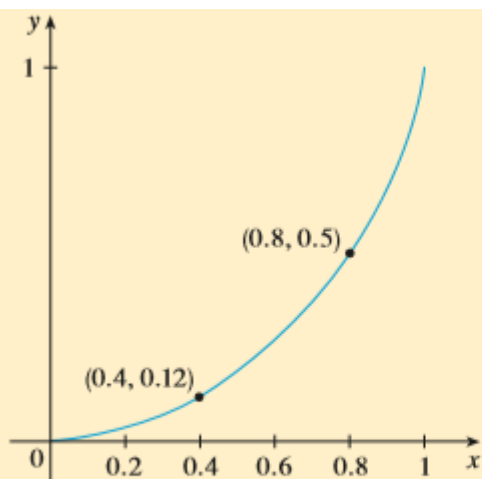


FIGURE 1

Lorenz curve for the US in 2008

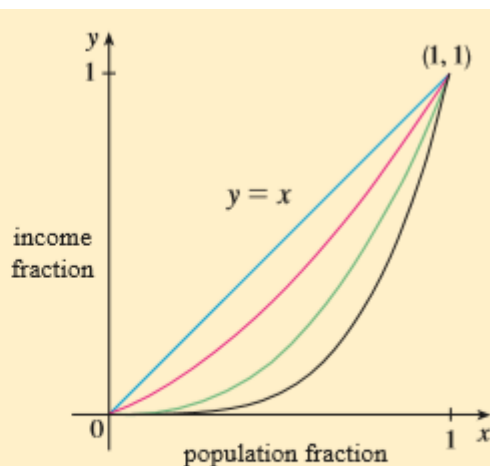


FIGURE 2

